Wikipedia Edit Histories: How is Truth or Belief "Drafted"?

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Abstract

As the world's largest collaborative knowledge platform, Wikipedia is where many people go to seek out truth. However, over the years, the site has been thought to be unreliable due to its crowdsourced nature and the possibility of users posting false information. This study investigates how temporal factors, like the edit rate over time, can be used to determine when an article can be trusted. To do so, we analyzed data using the MediaWiki API over a number of articles to look for specific trends, with the focus of our study being on two distinct groups of articles: scientific articles (e.g., Artemis II) and controversial articles (e.g., Donald Trump).

Our findings support that the moments when the edit rate is highest are when an article is most likely to have misinformation or vandalism. In both types of articles, we saw that around an event there was a spike in edit rates, with controversial articles usually having a secondary or tertiary spike. We used a variety of methods to determine whether these indicated increases in vandalism, such as sentiment analysis of the edit summaries and looking for specific phrases to identify periods of distrust.

However, when combing through metadata, we found that only when there was an increase in edit rates was there an increase in edit summaries indicating vandalism, ultimately proving that Wikipedia articles generally can not be trusted around an event. With those spikes in revisions either indicating vandalism or correcting of information with either way making the article not reliable.

Introduction:

Wikipedia is an internet encyclopedia where anyone can make an edit to an article. As such, it is able to collect information quickly and from a wide range of sources as many are able to edit an article at once, making it a collection of information from all across the internet. Therefore, it is an incredibly important resource for digital information consumption as it often is a collection of everything there is to learn about a topic.

In the modern day, Wikipedia is often the default baseline for factual knowledge to many. However, its open-access, crowdsourced architecture has led many to question its reliability, especially as the world becomes more polarized.

In our study, our goal is to determine when this polarization begins to affect an article or if it does at all through the lens of edit rates with the assumption that a high edit rate means either consensus being formed or vandalism. To accomplish this, we looked at a number of articles we believed would be either scientific and ones that would be controversial. These assumptions were done at our discretion with our logic being that articles around public policy and governance would be controversial while articles around technological milestones and empirical data would be scientific.

Related Work

Wikipedia's collaborative editing process has been extensively studied to understand how communities collectively create and maintain knowledge. Much of the prior research has focused on edit wars, which are conflicts where editors repeatedly override each other's changes. For example, Yasseri et al. (2014) examined edit wars across various Wikipedia topics and found that persistent disagreements are especially common in socio-political articles. Similarly, Viégas et al. (2004) used history flow visualizations to identify characteristic zigzag patterns that signify "edit wars", which are periods where two or more editors alternate changes over several consecutive revisions. Interestingly, they discovered that edit wars are not limited to highly controversial subjects but can also occur unexpectedly in less contentious pages, such as the article on "chocolate." In this study, we define a "raid" as a malicious edit war, resulting from a group or a party's coordinated effort to vandalize or disrupt a Wikipedia article. Such raids

often involve multiple users acting together to overwhelm the page with damaging edits.

Given the disruptive nature of edit wars and raids, we question whether these conflicts can be predicted or detected early to help enforce timely interventions. Wikipedia already employs a robust system of community norms and automated tools to regulate content quality. Halfaker et al. (2011) describe how Wikipedia has evolved from an open platform where anyone can edit to a more regulated environment, where contributors must understand social norms and navigate the website's algorithm (semi-automated rejections). While rejecting unwanted edits is the primary quality control mechanism, this approach has the unintended side effect of discouraging some well-meaning newcomers. Instead of outright rejection, some contributions that add value but don't fit perfectly are often rewritten or adjusted by editors. As for algorithms, although these tools have greatly improved efficiency in managing vandalism and disruptive behavior, they may inadvertently discourage newcomers. But, without the use of algorithms, far more volunteer effort would be needed to maintain Wikipedia's quality and integrity.

Our study builds upon this foundation by comparing the editing dynamics of socio-political topics with scientific and event-driven articles, such as Artemis II and 2026 in science. This allows us to investigate how truth-seeking behavior differs from ideological gridlock in different contexts. By applying techniques like TF-IDF, sentiment analysis, and Raid Ratio timelines, our research aims to improve the detection of edit wars and raids. In doing so, we hope to inform better algorithms and reduce automatic rejections of valuable edits, thereby fostering a more inclusive and constructive editing environment on Wikipedia.

Dataset

As stated previously we have the entirety of Wikipedia as the initial dataset available to us through the MediaWiki API. However, analyzing that many articles is an impossible task so we decided to focus on articles we had some knowledge of so we could reasonably assume whether they were controversial or not.

With that as our goal we eventually settled on a number of topics we believed would properly fall into these categories. Firstly for controversial we decide to look at the 2026 Iran war and Donald trump around the 2020 election. We decided on these topics because we wanted one topics we would be familiar with and know when events happened regarding them and two because they are either social, political or otherwise fear-inducing topics which we believed would spark vandalism.

Next, as our baseline we wanted non or less controversial topics. After consideration we decided on Artemis II, 2026 in science, and Donald Trump around the 2024 election. We first chose Artemis II as we believed it to be a purely scientific event based article that would only show updates around major events. Then we chose Trump around the 2024 election as it was less controversial than the 2020 election and therefore could be used as a point of comparison against it. We additionally looked at the Hantavirus page as it is a mix of both controversial and scientific. With our reasoning being that it has a scientific core but has a fear inducing element as people were afraid of another COVID-like virus.

From those articles, we focused on a narrow range of variables that we wanted to focus on in the analysis. For each article in the specified time frame we pulled timestamp of edit, comment metadata, if the edit was flagged minor or major, and finally we could optionally pull the actual edit itself. We decided on this narrow set of variables to maintain our project scope around edit rates which we could configure around the timestamp variable. We also chose to put more focus on the comment meta data than text data as it is shorter and easier to process than text edits that can span the entire article. Comment meta data is also more to the point and gives specific information on why someone performed an action as well as giving us the sentiment behind an edit. Finally we included the minor and major flag so we could discriminate between a content addition and a fixing of grammar.

Methods

We developed a data pipeline to collect and analyze data from Wikipedia's MediaWiki Action API. At its

core is a Python function, get_refined_history, which extracts historical data about Wikipedia articles. This function requires four arguments:

- title: the name of the Wikipedia article
- start_date: the beginning of date for retrieving edits
- end_date: the ending date for retrieving edits
- include_content: a boolean flag to determine whether the edit content should be included

The function sends requests to Wikipedia's API under the arguments listed above to retrieve data in JSON format. It also includes a HTTP header, User-Agent, which identifies the software making the request to provide more context for Wikipedia's API. In this case, the User-Agent is set to EditVisualizerBot/1.0 and includes a contact email from one of our members to comply with Wikipedia's API guidelines. The function retrieved data in batches (using the revision limit parameter) and organized the results in chronological order (oldest to newest). For each edit (called a "revision"), it collected key details like:

- Timestamp (exact time of the edit)
- Edit comments (notes left by users explaining their changes)
- Edit type flags (whether the edit was marked as minor or major by Wikipedia).

At first, we tried to extract the content of every edit by setting include_content=true. However, every edit encompasses the entire Wikipedia page content-wise. As a result, this created too much unnecessary data (noise), making it hard to focus on studying the edit rates and the sentiment/ keywords that potentially signal raids. So, we set include_content=false and focused only on the edit comments, which are the user-submitted notes, explaining each edit.
Once we collected the data, we first converted the raw data into a structured format using pandas DataFrames. We then cleaned the timestamps so they would be in a consistent format (datetime objects) for accurate analysis. Next, we categorized the edits into two groups based on Wikipedia's internal flags:

- Major edits: edits flagged as minor = false
- Minor edits: edits flagged as minor = true

To identify periods of high activity, we grouped the data by month and pinpointed the top three months with the highest edit density for each topic. We then analyzed these peak periods in greater detail at a daily frequency, calculating:

- The exact daily edit rates
- The ratio of major to minor edits during these active periods

The edit comments, which again are user-submitted notes explaining changes made, were unstructured text. To prepare this data for analysis, we removed potential formatting issues by:

- Decoding double-escaped HTML entities
- Eliminating non-breaking spaces and Wikipedia's internal section markers
- Stripping out hyperlinks and special punctuation noise

We then tokenized the cleaned text by:

- Splitting the text into individual words
- Converting all words to lowercase
- Filtering out English stop words (e.g., "the," "and," "a") using the NLTK corpus

To rank the importance of words in the dataset, we used TF-IDF (Term Frequency-Inverse Document Frequency). Based on what we learned in class, the statistical measure evaluates how important a word is to a document in a collection or corpus. In our case:

- Term Frequency (TF): the number of times a word appears in an edit comment divided by the total number of terms in that comment
- Inverse Document Frequency (IDF): the natural logarithm of the total number of comments divided by the number of comments containing the word

As a result, The TF-IDF score for each word was calculated as: TF-IDF = (Term Frequency) × log(Total Comments / Comments with the Term). We further analyzed the edit comments to uncover emotional and thematic patterns.

Using NLTK's VADER lexicon, we scored the sentiment of major edit comments. This lexicon categorizes comments as positive, negative, or neutral based on their compound scores, providing insight into the emotional tone of edits.

We trained an unsupervised LDA (Latent Dirichlet Allocation) model using the gensim library to

discover hidden themes in editor discussions. The model ran for 10 passes and 50 iterations, grouping similar edit comments into three main themes that represent the latent topics driving editor debates.

To identify conflicts and controversial editing patterns, we selected a list of raid keywords, including “revert,” “undid,” “rvv”, “vandal,” “hoax”, “attack”. Using the keywords, we built a daily “Raid Ratio” metric to track active edit wars as a percentage of total daily edits. This metric helped us identify conflicting editing patterns, i.e. raids, in real time.

Results and Discussion

Our analysis of long-term edit timelines revealed a notable behavioral contrast between scientific articles and controversial or conflict-driven pages. Political topics, for instance, exhibit high edit activity that has persisted for over two decades. This pattern is often interrupted by event-driven traffic spikes that are tied to global developments. For example, the Wikipedia page for Donald Trump demonstrated much cyclical volatility, with peaks aligning closely with major political events. Historical maxima were observed in November 2024 (1,545 edits), November 2016 (1,475 edits), and January 2025 (1,187 edits).

In contrast, science topics, for example, tend to undergo extended periods of minimal or no activity, before experiencing sudden, isolated explosions of revisions. This pattern was evident in the Artemis II Wikipedia page, which remained flat for years before surging to 1,109 edits in April 2026 as public interest in the recent lunar mission peaked.

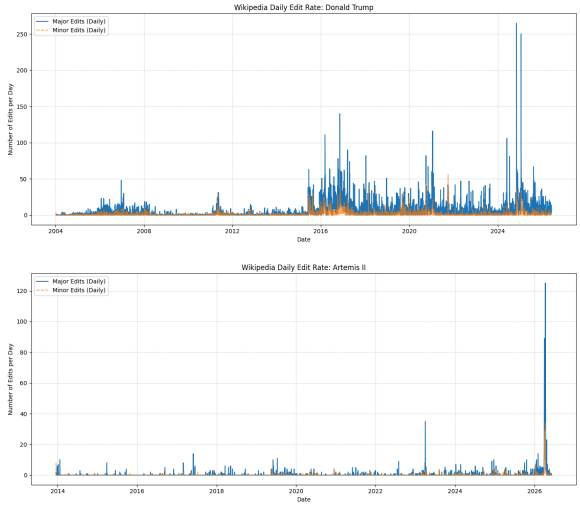


Figure 1. Long-term Wikipedia edit rate timelines across political and scientific articles. The Donald Trump page (top) demonstrates persistent cyclical macro-volatility aligned with election cycles, while the Artemis II page (bottom) exhibits prolonged dormancy punctuated by a milestone-driven surge.

Our review of these peak activity periods showed that non-minor revisions dominated the total traffic volume across most articles studied. In November 2024, non-minor edits accounted for 88.9% of the Donald Trump page’s monthly traffic. In April 2026, non-minor edits made up 78.7% of activity on the Artemis II page. Similarly, for the 2026 Iran War page in March 2026, non-minor edits represented 86.9% of total revisions.

This pattern suggests that peak edit periods were not driven only by small cleanup edits, formatting changes, or spelling corrections. Instead, these spikes involved major revision activity, likely reflecting a mixture of content expansion, correction, source updates, and administrative responses to unstable information. The 2026 in science page showed a comparatively smaller non-minor share, with 35.8% major edits, suggesting that its editing activity was more evenly divided between substantial additions and smaller maintenance edits.

To gain deeper insights, we zoomed in on peak months. Our analysis revealed three distinct editorial trajectories across the articles:

1. Rapid Decay Trajectory: The 2026 Iran War page experienced an unusually large surge in activity at the beginning of March 2026, with daily non-minor edit counts reaching roughly 800 to 900 edits around March 1-2. Activity then declined sharply over the following two weeks. This pattern is consistent with an initial crowdsourced effort to rapidly build, correct, and stabilize a breaking-news article, followed by a slow decline as the page became more settled.
2. Continuous Volatility: The Donald Trump page demonstrated a persistent and erratic edit pattern, with a major post-election surge on November 5, 2024. After this spike, the page continued to experience recurring secondary bursts of activity, often reaching

roughly 50 to 100 non-minor edits every few days. This volatility reflects the page's sensitivity to fast-moving political events and continuing public attention.

3. Event-Driven Spike: The Artemis II page shows elevated activity from the start of the month, followed by a larger spike of more than 120 non-minor edits around April 10. This pattern suggests event-driven updating around the Artemis II mission.

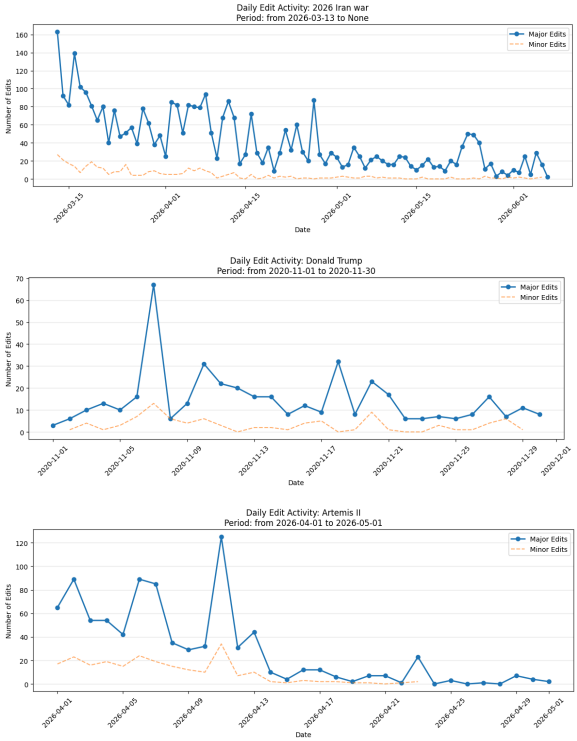


Figure 2.Micro-scale daily edit volume trajectories within peak months.

Our text mining efforts helped identify several recurring themes in the edit comments. The three-topic LDA model suggested three broad thematic clusters:

- Topic 1 (Technical Page Maintenance): This topic included high-probability terms such as added, reference, altered, name, removed, website, ref, and title. These terms suggest edits related to source updates, formatting, article structure, and routine maintenance.
- Topic 2 (Real-World Geopolitical Updates): This topic was dominated by terms such as iran, trump, iranian, war, military, and israel. These words suggest that edit-comments were tied to real-world developments.
- Topic 3 (Administrative Policy Enforcement): This topic included terms such as talk, source, article, revision, per, undid, section, and sentence. These terms point toward policy enforcement, content disputes, source debates, and administrative cleanup. Because LDA is exploratory, these topics should be interpreted as broad patterns rather than fixed categories.

Our sentiment analysis and vandalism tracking provided quantitative insights into the active conflict levels on these pages. The VADER classifications revealed that the vast majority of major edit comments remained objective and neutral, with neutral shares ranging from 61.4% on the Donald Trump page to 84.2% on the Hantavirus page.

A deeper dive into the sentiment results showed that the most negative comment cluster had an average VADER compound score of -0.1266 and a negative comment share of 35.96%. However, this negativity should not automatically be interpreted as toxic behavior between editors. Many negative scores appear to come from factual conflict-related terms such as attack, war, strike, and kill, which are expected in comments about military or geopolitical events. Therefore, the sentiment results are useful as a signal of emotionally or conflict-heavy language, but they do not directly prove editor conflict.

The Raid Ratio timelines suggest that scientific or event-driven pages such as Artemis II experienced relatively low levels of conflict-related edit comments, with only small, isolated spikes. In contrast, conflict-related pages such as the 2026 Iran war page showed more frequent use of administrative or revert-related language, suggesting a high level of conflict. The Raid Ratio, however, should be interpreted carefully because keywords such as "revert," "undid," or "vandal" capture administrative cleanup language, not necessarily vandalism.

We also observed that secondary spikes sometimes had higher conflict-related ratios than the initial event spike, especially for controversial articles. This may indicate that early edits focused on rapidly adding

information, while later edits involved correction, source disputes, or vandalism cleanup. Therefore, periods of peak popularity should be treated as moments of heightened uncertainty rather than automatic proof that the article is unreliable.

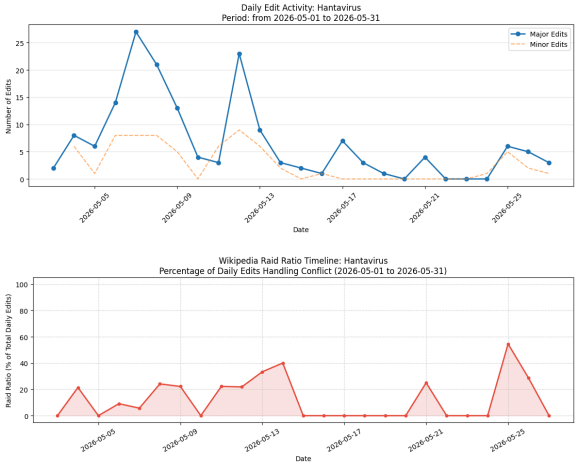


Figure 3.Temporal decoupling of edit volume and community conflict on the Hantavirus page

Ethical Considerations

While our project relies entirely on public data from the Wikipedia API, analyzing community behavior still requires several ethical safeguards.

Firstly, we need to take into consideration user privacy and anonymity. Even though edit histories are public, mapping out editor volume and tracking specific comment histories could inadvertently spotlight individual contributors. For editors living in volatile regions or editing highly sensitive topics (like the Iran War), preserving anonymity is critical to ensure their online contributions do not lead to real-world consequences.

Another thing is sentiment analysis bias. Tools like VADER are trained on social media text and can easily misinterpret Wikipedia's procedural language. For example, an administrative comment like "reverted vandalism" might be flagged as negative sentiment, which risks mischaracterizing hard-working moderators as toxic or hostile.

There's also a probability of misidentiying bots as human. High edit rates and identical text similarity can look like intense arguments among humans, but they are sometimes just automated scripts or bots doing standard page maintenance. We must be careful not to label automated maintenance as genuine human conflict.

Conclusion

In conclusion, our analysis shows that Wikipedia edit patterns look very different depending on the topic. Scientific articles usually stay quiet until a major milestone happens, while political and controversial articles are constantly active and prone to massive spikes in edits when news breaks.

While the Wikipedia community works hard to maintain these pages - mostly through thousands of minor cleanup edits - the data tells a clear story about reliability. When a major event happens and a page experiences a massive spike in traffic, it is highly likely that the page is experiencing vandalism or undergoing heavy corrections. Because of these intense "edit wars" and secondary spikes in revisions, our findings ultimately prove that during moments of peak public attention, Wikipedia articles are at their most unreliable and should not be fully trusted until the activity settles down.

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